Quartic normal equations: complete-space solvability, sharp inverse stability, and atomic obstructions

Project Stardust

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Abstract

Let H be a real Banach space and let $\delta : H \to L^4(\mu)$ be bounded linear with $c\,\|u\| \le \|\delta u\|_4$ for some $c > 0$. The quartic normal map $\mathcal{A}(u)(v) = \int (\delta u)^3 \delta v \, d\mu$ is a bijection from H onto its continuous dual. We give a direct proof using norm-Cauchy minimizing sequences and obtain a global inverse estimate with exponent $1/3$. A weighted lower bound at a source gives a stronger linear stability estimate relative to that source. An actual infinite Gaussian first-chaos embedding has a radial normal map and regular inverse at every nonzero datum; its sharp fixed-source estimate is derived explicitly. On infinite Poisson L^4, however, the exponent $1/3$ is locally optimal at every dual datum, including data whose inverse is nonzero almost everywhere. Every fixed finite-dimensional set of source directions can still admit linear stability at such a source. The results are reconstructed from a frozen Lean development; the existence and uniform-convexity mechanisms are classical. The purpose is a complete account of the estimates, their hypotheses, and the distinction between finite-direction and ambient regularity. Priority for the combined presentation remains unresolved.

1 The equation and its regularity questions

Fix a measure space (Ω, Σ, μ). No finiteness or σ-finiteness assumption is imposed unless explicitly stated. All L^p spaces in this paper consist of equivalence classes of real measurable functions modulo equality almost everywhere. Let H be a real Banach space with norm $\|\cdot\|$ and continuous dual H^*, equipped with its operator norm. Suppose that $\delta : H \to L^4(\mu)$ is bounded and linear and that

$$c\,\|u\| \le \|\delta u\|_4 \qquad (u \in H), \qquad c > 0. \tag{1}$$

Write $J(F) = F^3 \in L^{4/3}(\mu)$ and define

$$\mathcal{A}(u)(v) = \int_\Omega (\delta u)^3 \delta v \, d\mu, \qquad u, v \in H. \tag{2}$$

The notation $\delta^* J(\delta u)$ is permissible if δ^* means the transpose of δ applied to the integral functional represented by $J(\delta u)$. We do not identify all of $(L^4)^*$ with $L^{4/3}$ as a preliminary assumption. This matters on general measure spaces.

The lower bound in (1) is quantitative. Injectivity alone does not replace it in infinite dimension. It makes $u \mapsto \|\delta u\|_4$ an equivalent norm and makes the image of δ closed. Indeed, if δu_n converges in L^4, the lower bound makes u_n Cauchy, and completeness and continuity identify the limit. This elementary observation explains why the source norm is controlled throughout the paper.

Theorem 1.1. *Under* (1)*, the map* $\mathcal{A} : H \to H^*$ *is a smooth bijection. Its inverse* T *satisfies, for all* $f, g \in H^*$*,*

$$\|Tf - Tg\|^3 \le \frac{4}{c^4} \|f - g\| . \tag{3}$$

If H *contains a nonzero vector, no exponent* $q > 1/3$ *gives a local Hölder estimate for* T *at zero. If* $H = L^4(\mu_\lambda)$*,* δ *is the identity, and* μ_λ *is the Poisson law of a parameter* $\lambda > 0$*, no such exponent gives a local estimate at* any *datum of* H^**.*

The final assertion is stronger than failure of a uniform global Lipschitz constant. It excludes even an estimate comparing one fixed datum to nearby data. Conversely, we will prove such a fixed-datum linear estimate under an additional weighted lower bound. Its constant need not work between two arbitrary nearby data.

The existence argument belongs to the classical geometry of uniformly convex spaces, initiated by Clarkson [1]. Monotone-operator solvability provides another classical context [2, 3]. We do not claim a new general surjectivity principle or a new uniform-convexity inequality. The direct proof is useful here because it exposes the precise quantitative input and avoids a separate reflexivity construction. In classical analysis the closed embedding in L^4 already implies reflexivity; avoiding that theorem in the proof does not produce nonreflexive examples satisfying (1).

2 The actual duality map and its energy

Hölder's inequality gives

$$\|J(F)\|_{4/3} = \|F\|_4^3 , \qquad \int J(F) F \, d\mu = \|F\|_4^4 . \tag{4}$$

For example, $\int |F^3|^{4/3} = \int |F|^4 < \infty$, which proves membership and the norm identity directly. Almost-everywhere equality of two representatives implies the same equality for their cubes, so J is well defined on the quotient space. The product $J(F)G$ is integrable for every $G \in L^4$. Consequently (2) defines a continuous functional, with

$$\|\mathcal{A}(u)\| \le \|\delta\| \, \|\delta u\|_4^3 , \qquad \langle \mathcal{A}(u), u \rangle = \|\delta u\|_4^4 . \tag{5}$$

Multiplication is a bounded bilinear map $L^4 \times L^4 \to L^2$ and also $L^2 \times L^4 \to L^{4/3}$, both of norm at most one. The cube map is the composition of these maps along the diagonal. It is therefore a continuous cubic polynomial, in particular smooth. Composition with δ and the bounded transpose pairing proves smoothness of $\mathcal{A}$. This argument concerns Fréchet differentiability in the specified Banach norms; pointwise differentiation alone would not establish it.

For $f \in H^*$ put

$$E_f(u) = \frac{1}{4} \|\delta u\|_4^4 - f(u). \tag{6}$$

Lemma 2.1. *The energy is continuously Fréchet differentiable, and* $DE_f(u) = \mathcal{A}(u) - f$*.*

Proof. Write $F = \delta u$ and $G = \delta h$. Expansion of the quartic yields

$$\frac{1}{4} \int (F + G)^4 - \frac{1}{4} \int F^4 - \int F^3 G = \frac{3}{2} \int F^2 G^2 + \int F G^3 + \frac{1}{4} \int G^4 .$$

Every integral is absolutely convergent by Hölder's inequality. The absolute value of the right side is at most

$$\frac{3}{2} \|F\|_4^2 \, \|G\|_4^2 + \|F\|_4 \, \|G\|_4^3 + \frac{1}{4} \|G\|_4^4 .$$

Since $\|G\|_4 \le \|\delta\|\,\|h\|$, this is $o(\|h\|)$. The linear term is exactly $\mathcal{A}(u)(h)$. Smoothness of $\mathcal{A}$, established above, gives continuity of the derivative. □

Lemma 2.2. *The range of E_f is bounded below. More precisely, with $R = \max\{1, 4\|f\|/c^4\}$, one has $E_f(u) \ge -\|f\|\,R$.*

Proof. If $E_f(u) \le 0$ and $s = \|u\|$, then $c^4 s^4 \le 4\|f\|\,s$. For $s > 1$ this implies $c^4 s \le 4\|f\|$, hence $s \le R$; for $s \le 1$ the bound is immediate. Thus $E_f(u) \ge -f(u) \ge -\|f\|\,R$ on the zero sublevel. Outside that sublevel the same lower bound holds because the energy is positive and $\|f\|\,R \ge 0$. □

3 Minimization without weak compactness

The following elementary lemma isolates the whole existence mechanism. It will be applied after the required deficit has been proved for the actual quartic integral.

Lemma 3.1 (Quartic midpoint minimization)**.** *Let X be a real Banach space and let $E : X \to \mathbb{R}$ be continuous and bounded below. If $k > 0$ and*

$$E\left(\frac{x+y}{2}\right) + k\,\|x-y\|^4 \le \frac{E(x)+E(y)}{2} \qquad (x, y \in X), \tag{7}$$

then E attains its infimum.

Proof. Let $\alpha = \inf E(X)$, a finite real number. Choose x_n with $E(x_n) \to \alpha$. Since $\alpha \le E((x_n + x_m)/2)$, (7) implies

$$k\,\|x_n - x_m\|^4 \le \frac{E(x_n) - \alpha + E(x_m) - \alpha}{2}.$$

Given $\varepsilon > 0$, eventually both energy differences are less than $k\varepsilon^4$. The displayed inequality then gives $\|x_n - x_m\| < \varepsilon$. Completeness supplies a norm limit x, and continuity gives $E(x) = \alpha$. □

Lemma 3.2. *The energy* (6) *satisfies*

$$E_f\left(\frac{u+v}{2}\right) + \frac{c^4}{64}\,\|u-v\|^4 \le \frac{E_f(u)+E_f(v)}{2}. \tag{8}$$

Proof. For real a, b, direct expansion gives

$$\frac{a^4+b^4}{2} - \left(\frac{a+b}{2}\right)^4 - \frac{(a-b)^4}{16} = \frac{3}{8}(a^2-b^2)^2 \ge 0.$$

Substitute representatives of δu and δv and integrate. All fourth powers are integrable, so the resulting inequality is valid for the quotient classes. Multiplication by $1/4$ gives the deficit $\|\delta(u-v)\|_4^4/64$ for the potential. The forcing term cancels exactly at the midpoint by linearity of f. Finally apply (1) to $u - v$. □

Proposition 3.3. *For each $f \in H^*$ there exists $u \in H$ with $\mathcal{A}(u) = f$.*

Proof. Lemmas 2.2, 3.2, and 3.1 produce a global minimum u of E_f. For any $h \in H$, the real function $s \mapsto E_f(u + sh)$ has a minimum at zero, so its derivative there vanishes. Lemma 2.1 identifies that derivative with $(\mathcal{A}(u) - f)(h)$. Since this holds for every h, $\mathcal{A}(u) = f$. □

The proof uses neither compactness of a norm ball nor an unproved weakly convergent subsequence. The distinction is useful in a formal development: the only limiting step is completeness applied to a sequence already proved Cauchy. The argument also covers the zero space, where the equation has its unique trivial solution.

4 Uniform monotonicity and global inverse stability

Existence did not need a uniqueness assumption. The same scalar quartic geometry supplies uniqueness and the precise modulus of the inverse.

Lemma 4.1. *For $u, v \in H$,*

$$\langle \mathcal{A}(u) - \mathcal{A}(v), u - v \rangle \geq \frac{1}{4} \|\delta(u - v)\|_4^4 \geq \frac{c^4}{4} \|u - v\|^4 . \tag{9}$$

Proof. The scalar identity

$$(a^3 - b^3)(a - b) - \frac{1}{4}(a - b)^4 = \frac{3}{4}(a - b)^2(a + b)^2$$

has a nonnegative right side. Its terms are integrable after substituting $a = \delta u$, $b = \delta v$, by the $L^{4/3}$–L^4 pairing and quartic integrability. Integration gives the first inequality, and (1) gives the second. □

Proof of bijectivity and (3). If $\mathcal{A}(u) = \mathcal{A}(v)$, (9) forces $u = v$. Together with Proposition 3.3, this defines the inverse $T : H^* \to H$. For $u = Tf$, $v = Tg$, dual norm control gives

$$\frac{c^4}{4} \|u - v\|^4 \leq (f - g)(u - v) \leq \|f - g\| \, \|u - v\| .$$

When $u \neq v$, divide by $\|u - v\|$; otherwise the desired estimate is immediate. Taking cube roots gives the global Hölder bound

$$\|Tf - Tg\| \leq (4/c^4)^{1/3} \, \|f - g\|^{1/3} .$$

□

Proposition 4.2. *Suppose $u \neq 0$. For every $q > 1/3$, $K \geq 0$, and $r > 0$, there exists $f \in H^*$ with $\|f\| < r$ and*

$$\|Tf - T0\| > K \, \|f\|^q .$$

Proof. The defining integral gives $\mathcal{A}(au) = a^3 \mathcal{A}(u)$ for real a and $\mathcal{A}(0) = 0$. Put $f_a = \mathcal{A}(au)$ for $a > 0$. Then $Tf_a = au$, $T0 = 0$, and $\|f_a\| = a^3 \, \|\mathcal{A}(u)\| \to 0$. The proposed estimate would require

$$\|u\| \leq K \, \|\mathcal{A}(u)\|^q \, a^{3q-1}.$$

The right side tends to zero, whereas $\|u\| > 0$. Choosing a small enough gives the strict reverse inequality and $\|f_a\| < r$. □

The exponent is optimal in this precise local sense; no claim that the global constant is optimal for every δ is needed. The singularity at zero follows from cubic homogeneity. To understand nonzero data we need information about multiplication by the source.

5 Weighted source control and finite-dimensional directions

For $u \in H$, define the bounded linear map

$$\mathcal{B}_u : H \to L^2(\mu), \qquad \mathcal{B}_u w = (\delta u)(\delta w). \tag{10}$$

Hölder's inequality gives $\|\mathcal{B}_u w\|_2 \le \|\delta u\|_4 \, \|\delta\| \, \|w\|$. This is an actual L^2 product, with equality of representatives understood almost everywhere.

Lemma 5.1. *For all $u, v \in H$,*

$$\langle \mathcal{A}(u) - \mathcal{A}(v), u - v \rangle \ge \frac{3}{4} \, \|\mathcal{B}_u(u - v)\|_2^2 . \tag{11}$$

Proof. For real a, b,

$$(a^3 - b^3)(a - b) - \frac{3}{4} a^2 (a - b)^2 = (a - b)^2 (b + a/2)^2 \ge 0.$$

Integration gives the assertion. The square of the product is integrable because $(\delta u)(\delta(u - v)) \in L^2$. □

Theorem 5.2 (Stability relative to one datum)**.** *If $\beta > 0$ and $\|\mathcal{B}_u w\|_2 \ge \beta \, \|w\|$ for every $w \in H$, then, for every $v \in H$,*

$$\|u - v\| \le \frac{4}{3\beta^2} \, \|\mathcal{A}(u) - \mathcal{A}(v)\| . \tag{12}$$

Under (1)*, setting $f_0 = \mathcal{A}(u)$ yields the corresponding bound $\|T f_0 - T f\| \le 4 \, \|f_0 - f\| / (3\beta^2)$ for all $f \in H^*$.*

Proof. By Lemma 5.1 and the dual norm inequality,

$$\frac{3}{4} \beta^2 \, \|u - v\|^2 \le \|\mathcal{A}(u) - \mathcal{A}(v)\| \, \|u - v\| .$$

Divide by the distance when it is positive. The zero case is immediate. □

Proposition 5.3. *Suppose δ is injective and δu is nonzero almost everywhere. For each linear subspace $E \subset H$ of finite dimension, there is $K_E > 0$ such that*

$$\|u - v\| \le K_E \, \|\mathcal{A}(u) - \mathcal{A}(v)\| \qquad \text{whenever } u - v \in E. \tag{13}$$

In particular, if H is finite dimensional the estimate holds for every $v \in H$.

Proof. If $\mathcal{B}_u w = 0$, then $(\delta u)(\delta w) = 0$ almost everywhere. Nonvanishing of δu gives $\delta w = 0$, and injectivity gives $w = 0$. Thus $\mathcal{B}_u$ is injective. Restricted to E, it has a positive lower norm bound: for $E \neq \{0\}$, minimize the continuous positive function $w \mapsto \|\mathcal{B}_u w\|_2$ on the compact unit sphere of E. Its minimum β_E is positive; scaling gives the bound on E. For $E = \{0\}$ any positive lower-bound constant works. Apply the proof of Theorem 5.2 to $w = u - v \in E$. □

The finite-dimensional constant depends on the subspace. The proposition provides no uniform lower bound as E varies. This dependence is decisive on an infinite atomic space.

As a regular example, let μ be a probability measure, $H = \mathbb{R}$, and δu the constant function u. Then $\|\delta u\|_4 = |u|$ and $\|\mathcal{B}_u w\|_2 = |u||w|$. At $u \neq 0$, (12) gives a linear stability constant $4/(3|u|^2)$. Under the usual identification of $\mathbb{R}^*$ with $\mathbb{R}$, the normal equation is $u^3 = f$. The behavior at zero and at nonzero data is thus already different in one dimension.

6 Gaussian synthesis and inverse regularity away from zero

The infinite-dimensional failure below is not a consequence of dimension alone. Consider the actual Gaussian product space carrying independent standard coordinates $(G_j)_{j\ge 0}$, and take $H = \ell^2(\mathbb{N};\mathbb{R})$. Its isonormal embedding $W(h)$ is the L^2 limit of $\sum_{j<N} h_j G_j$. Every finite linear combination of such limits is Gaussian: convergence in L^2 implies convergence of characteristic functions, whose finite formulas tend to $\exp(-\|h\|^2 t^2/2)$. In particular

$$\mathbb{E}W(h)^4 = 3\|h\|^4, \qquad \delta h := W(h) \in L^4, \qquad \|\delta h\|_4 = 3^{1/4}\|h\|.$$

Linearity holds on the actual equivalence classes: it first holds in L^2, hence almost everywhere, and the representatives belong to L^4. Thus $\delta : H \to L^4$ is a bounded closed embedding satisfying (1). It is not the identity on the whole Gaussian L^4.

For $h, k \in H$, the joint moment generating function is $\exp((\|h\|^2 s^2 + 2\langle h,k\rangle st + \|k\|^2 t^2)/2)$. Its s^2t^2 and s^3t coefficients give

$$\begin{aligned} \|\mathcal{B}_h k\|_2^2 &= \mathbb{E}[W(h)^2 W(k)^2] = \|h\|^2\|k\|^2 + 2\langle h,k\rangle^2, \\ \mathcal{A}(h)(k) &= \mathbb{E}[W(h)^3 W(k)] = 3\|h\|^2\langle h,k\rangle. \end{aligned} \tag{14}$$

These are genuine finite moments by Gaussian integrability and Hölder's inequality, including degenerate source pairs. Hence $\|\mathcal{B}_h k\|_2 \ge \|h\|\|k\|$ on the *entire* infinite source space. At every $h \neq 0$, Theorem 5.2 already gives the mapped fixed-datum estimate

$$\|h - Tf\| \le \frac{4}{3\|h\|^2}\|\mathcal{A}(h) - f\|. \tag{15}$$

The source `GaussianFirstChaosQuarticSource` proves the actual L^4 realization, these two moment identities and this estimate.

6.1 Exact radial inverse and a sharp fixed-datum constant

Identify H^* with H by the real Riesz isometry $Rh(k) = \langle h,k\rangle$. Equation (14) identifies the normal map with

$$N(h) = 3\|h\|^2 h.$$

Its inverse, in these Hilbert coordinates, is

$$T(y) = \begin{cases} 3^{-1/3}\|y\|^{-2/3} y, & y \neq 0, \\ 0, & y = 0. \end{cases} \tag{16}$$

Indeed $\|N(h)\| = 3\|h\|^3$, and substitution checks both inverse identities. In particular the origin retains the cubic-root singularity. The following sharper bound and its optimality are explicit human consequences of the mapped radial identity, not additional named formal declarations.

Proposition 6.1. *For every fixed $h \neq 0$ and every $k \in H$,*

$$\|h - k\| \le \frac{4}{9\|h\|^2}\|N(h) - N(k)\|. \tag{17}$$

The constant is best possible for comparison with this fixed source against all other sources, already on its one-dimensional span. Equivalently it bounds $\|T(N(h)) - T(y)\|$ for every datum y.

Proof. Put $r = \|h\|$, $s = \|k\|$, $c = \langle h, k\rangle$ and $d = r^2 - rs + s^2$. Direct expansion of squared Hilbert norms gives

$$\begin{aligned}\|r^2 h - s^2 k\|^2 - d^2\|h - k\|^2 \\ = 2(r^2 + s^2)(r - s)^2(c + rs) \geq 0.\end{aligned}$$

The sign uses Cauchy–Schwarz, $c \geq -rs$. Both norms and d are nonnegative, so taking square roots proves $\|r^2 h - s^2 k\| \geq d\|h - k\|$. Moreover $d = (s - r/2)^2 + 3r^2/4 \geq 3r^2/4$. Multiplication by three gives (17). For $k = -h/2$, the source distance is $3r/2$ and the image distance is $27r^3/8$. Their ratio is $4/(9r^2)$, proving optimality. □

6.2 Local derivatives and the distinction between estimates

Away from zero, the scalar norm powers in (16) are continuously differentiable. Since $D(\|y\|^2)[v] = 2\langle y, v\rangle$, the scalar power rule and product rule give, at $y = N(h)$,

$$DT(y) = \frac{1}{3\|h\|^2}\left(I - \frac{2}{3}\Pi_h\right), \qquad \Pi_h v = \frac{\langle h, v\rangle}{\|h\|^2} h. \tag{18}$$

The derivative acts by $1/(9\|h\|^2)$ along h, and by $1/(3\|h\|^2)$ on its orthogonal complement. In this infinite Hilbert space the latter complement is nonzero, so the derivative norm is exactly $1/(3\|h\|^2)$. Formula (17) is a different, global comparison relative to a fixed nonzero datum; its sharp constant is larger than this local derivative norm.

For completeness, the inverse also has ordinary two-point local Lipschitz regularity. On a datum ball $\|y - y_0\| < \rho < \|y_0\|$, its derivative norm is bounded by $3^{-1/3}(\|y_0\| - \rho)^{-2/3}$. The segment joining two points of this ball stays in the ball. Integrating the continuous derivative along that segment gives the same bound times their distance. This is an elementary human differentiation of the explicit inverse, not an inference that the general one-base estimate alone implies an arbitrary two-point estimate.

The underlying radial map is the classical $p = 4$ vector power map; Lindqvist [4], Section 10, gives the standard vector power identities and monotonicity framework. The exact fixed-source constant above is proved directly here; the bounded comparison does not establish its priority in the literature. The contrast with the next section changes the source embedding: Gaussian first-chaos synthesis on ℓ^2 has uniform weighted control, whereas the full Poisson L^4 identity permits concentrated atomic perturbations. Neither infinite dimension nor nonvanishing of an individual source settles inverse regularity without that geometry.

6.3 Physical Gaussian functions do not supply the same embedding

The first-chaos synthesis above is a particular bounded operator into L^4, not an identity embedding of the whole scalar Gaussian space. Even adding the original first-derivative graph norm does not give the needed uniform bound. Let $\gamma = N(0, 1)$ and let D_γ be the closure in $L^2(\gamma)$ of differentiation on the polynomial core. For every real θ, the actual function $f_\theta(x) = e^{\theta x}$ belongs to both $L^2(\gamma)$ and $L^4(\gamma)$, with

$$\|f_\theta\|_2 = e^{\theta^2}, \qquad \|f_\theta\|_4 = e^{2\theta^2}.$$

These follow by completing the square in the Gaussian integral $\int e^{tx}\, d\gamma = e^{t^2/2}$. In particular their norm ratio e^{θ^2} is unbounded. A bounded linear operator from physical $L^2(\gamma)$ to $L^4(\gamma)$ agreeing

with these same functions on all exponentials cannot exist: its operator-norm inequality would bound that ratio uniformly.

The exponential also belongs to the *original* closed derivative graph, rather than only having a classical pointwise derivative. Indeed let $P_N(x) = \sum_{j=0}^{N} (\theta x)^j / j!$. Then $P_N \to f_\theta$ pointwise and $P_N' = \theta P_{N-1}$ for $N \geq 1$. The bounds $|P_N| \leq e^{|\theta x|}$ and $|P_N'| \leq |\theta| e^{|\theta x|}$ provide square-integrable Gaussian majorants, since $e^{2|\theta x|} \leq e^{2\theta x} + e^{-2\theta x}$. Dominated convergence gives convergence in both graph components. Closedness proves

$$f_\theta \in D(D_\gamma), \qquad D_\gamma f_\theta = \theta f_\theta.$$

Nevertheless

$$\frac{\|f_\theta\|_4}{\|f_\theta\|_2 + \|D_\gamma f_\theta\|_2} = \frac{e^{\theta^2}}{1 + |\theta|} \longrightarrow \infty.$$

Thus no constant controls the physical L^4 norm by this original graph norm, even on these common-domain analytic tests. The source also supplies a direct finite witness for each proposed real constant K, choosing $\theta = |K| + 1$ and using $e^{\theta^2} \geq 1 + \theta^2$. This is not a claim that any one displayed exponential fails L^4 integrability; all of them have finite moments of every finite order.

The comparison is consistent with the classical Gaussian logarithmic Sobolev inequality, whose output is entropy, not this L^4 estimate. For example the normalized test $g_\theta = e^{\theta x - \theta^2}$ has $\|g_\theta\|_2 = 1$ and, by direct Gaussian differentiation,

$$\int g_\theta^2 \log g_\theta^2 \, d\gamma = 2\theta^2 = 2\|D_\gamma g_\theta\|_2^2, \qquad \|g_\theta\|_4 = e^{\theta^2}.$$

These normalized identities are elementary human consequences of the same moment formula, not additional named formal declarations. Ledoux's primary author slides [5], PDF pages 7–8, state the classical Gaussian entropy inequality in this normalization. No new Gaussian embedding or entropy principle is claimed. To apply the quartic existence theorem one must prove its actual bounded L^4 source operator, as first-chaos synthesis does above; the physical scalar L^2 space and this graph norm do not supply that hypothesis automatically.

`GaussianL2QuarticEmbeddingObstruction`: actual exponential classes, exact norms and no bounded physical extension.

`GaussianExponentialSobolevObstruction`: Taylor graph approximation and absence of a graph-norm L^4 bound.

7 Infinite Poisson space and failure of weighted coercivity

Fix $\lambda > 0$. Give $\mathbb{N} = \{0, 1, \ldots\}$ its full power-set σ-algebra and Poisson probability measure

$$p_n = \mu_\lambda(\{n\}) = e^{-\lambda} \frac{\lambda^n}{n!} > 0.$$

In this section $H = L^4(\mu_\lambda)$ and δ is the identity. The lower bound (1) holds with $c = 1$. Every coordinate has positive mass, so an equivalence class has well-defined values $F(n)$. Its norm is

$$\|F\|_4^4 = \sum_{n=0}^{\infty} p_n |F(n)|^4.$$

Let $e_n = p_n^{-1/4} 1_{\{n\}}$. These are normalized in L^4. For a fixed $F \in H$ put $x_n = F(n) p_n^{1/4}$. Summability gives

$$\sum_n |x_n|^4 < \infty, \qquad x_n \longrightarrow 0. \tag{19}$$

Proposition 7.1. *No $F \in L^4(\mu_\lambda)$ satisfies a positive lower bound $\|Fw\|_2 \geq \beta \|w\|_4$ for all $w \in L^4(\mu_\lambda)$.*

Proof. Taking $w = e_n$ gives

$$\|Fe_n\|_2 = |F(n)|p_n^{1/4} = |x_n| \longrightarrow 0, \qquad \|e_n\|_4 = 1.$$

A positive lower bound contradicts this limit. □

Even the constant source $F = 1$ has this obstruction. It is nonzero everywhere, and multiplication by F is injective, but the inclusion of this probability-space L^4 into L^2 has no positive lower bound. This does not yet prove failure of inverse Lipschitz stability: Theorem 5.2 is a sufficient criterion, not an equivalence. We next prove the stronger failure directly by perturbing the normal map itself.

8 The inverse exponent is optimal at every atomic datum

The integral pairing represented by e_n^3 has dual norm at most one by Hölder's inequality. For $a \geq 0$, the functions $F + ae_n$ and F differ at only one coordinate. Cubing gives the exact identity

$$\mathcal{A}(F + ae_n) - \mathcal{A}(F) = \big((x_n + a)^3 - x_n^3\big)\mathcal{A}(e_n). \tag{20}$$

In particular,

$$\|\mathcal{A}(F + ae_n) - \mathcal{A}(F)\| \leq a\big(3|x_n|^2 + 3a|x_n| + a^2\big). \tag{21}$$

To check (20) without confusing values and normalized coordinates, observe that on the nth atom $F(n) = p_n^{-1/4}x_n$ and $e_n(n) = p_n^{-1/4}$. The cube difference is therefore $p_n^{-3/4}((x_n+a)^3 - x_n^3)$, exactly the multiplier of e_n^3. Off that atom both sides vanish.

Theorem 8.1. *For every $f_0 \in H^*$, $q > 1/3$, $K \geq 0$, and $r > 0$, there exists $f \in H^*$ such that*

$$\|f - f_0\| < r, \qquad \|Tf - Tf_0\| > K\|f - f_0\|^q. \tag{22}$$

Proof. By surjectivity write $F = Tf_0$. Let

$$a_n = |x_n| + \frac{1}{n+1}, \qquad U_n = F + a_n e_n, \qquad f_n = \mathcal{A}(U_n).$$

Then $a_n > 0$ and $a_n \to 0$ by (19). Since $|x_n| \leq a_n$, (21) gives

$$\|f_n - f_0\| \leq 7a_n^3 \longrightarrow 0.$$

Injectivity of $\mathcal{A}$ and $\|e_n\|_4 = 1$ give $\|Tf_n - Tf_0\| = \|U_n - F\|_4 = a_n$. Because $3q - 1 > 0$, eventually $K7^q a_n^{3q-1} < 1$. Hence

$$K\|f_n - f_0\|^q \leq K7^q a_n^{3q} < a_n.$$

Choose n large enough for this inequality and the radius condition. □

This completes the optimality assertion in Theorem 1.1. The proof uses small normalized coordinates of an arbitrary L^4 source, rather than a zero coordinate. Thus it applies to strictly positive sources as well as sources supported on finitely many atoms. The Poisson distribution is a concrete verified model; the argument itself only uses the displayed positive atomic weights and quartic summability. We do not claim a separate Lean theorem for every atomic measure under that generalization.

Corollary 8.2. *At the constant source $F = 1$, every fixed finite-dimensional source subspace $E \subset L^4(\mu_\lambda)$ admits* (13)*. There is no common constant valid for all one-dimensional source subspaces.*

Proof. Apply Proposition 5.3 to $\delta = \mathrm{id}$ and $F = 1$. If a constant K worked for all one-dimensional subspaces, then for each V take $E = \mathrm{span}\{1 - V\}$. The estimate would hold for every V. Taking $V = Tf$ contradicts Theorem 8.1 with $f_0 = \mathcal{A}(1)$ and $q = 1$. □

The order of quantifiers is the issue: for each fixed E there is a constant, but no single constant works for all E. A finite-dimensional calculation at a nonvanishing source can therefore be accurate while providing no uniform ambient regularity theorem.

9 Disjoint sources and the role of support

A simpler obstruction at selected nonzero data helps separate two mechanisms. Assume (1). Suppose $u, v \in H$, $v \neq 0$, and $(\delta u)(\delta v) = 0$ almost everywhere. Then mixed terms in the cubic vanish, so

$$\mathcal{A}(u + av) = \mathcal{A}(u) + a^3 \mathcal{A}(v) \qquad (a \in \mathbb{R}). \tag{23}$$

Proposition 9.1. *For every $K \geq 0$ and $r > 0$ there exists f with $\|f - \mathcal{A}(u)\| < r$ and $\|Tf - u\| > K \|f - \mathcal{A}(u)\|$.*

Proof. Use $f_a = \mathcal{A}(u + av)$ for $a > 0$. The source difference has norm $a \|v\|$, and (23) gives dual difference $a^3 \|\mathcal{A}(v)\|$. Small a meets both required inequalities. □

For the Poisson identity source one may take indicators of two distinct atoms. This produces a nonzero datum with no linear inverse estimate using disjoint support alone. Theorem 8.1 goes further: it works when the source never vanishes and no nonzero disjoint source exists. Small normalized tail coordinates, rather than exact support separation, then replace cubic decoupling.

10 Formal scope and reproducibility

The mathematical baseline is the frozen Git commit

`bab688bb6140cd6316c80680095a914ea0d346db.`

All declarations below have prefix `OperatorDerivativeRewrite.Foundation`. The accompanying `theorem-map.json` records exact files, declarations, and hashes. The principal correspondence is as follows.

Paper result	*Lean namespace and declaration*
(4), (2)	`BanachLpDualityFour.dualityFour_norm`, `dualityFour_pairing_eq_norm`, `normalFour`
Lemma 2.1	`BanachLpDualityFourExistence.hasFDerivAt_quarticPotential`
Lemmas 2.2–3.2	`BanachLpDualityFourCoercivity.quarticPotential_lower_bound`, `quarticEnergy_midpoint_lower`; `QuarticUniformMinimization.exists_minimizer_of_quartic_midpoint`
Bijectivity, (3)	`BanachLpDualityFourInfinite.normalFour_existsUnique`, `normalFour_inverse_holder`

Proposition 4.2	`BanachQuarticHolderSharpness.no_local_holder_bound`
Theorem 5.2	`BanachQuarticWeightedStability.inverse_lipschitz_at_weighted_datum`
Proposition 5.3	`BanachFiniteDirectionStability.finite_direction_stability`
Proposition 7.1	`PoissonQuarticWeightedObstruction.no_positive_weighted_lower`
(20), (21)	`PoissonQuarticAtomPerturbation.normal_atom_shift`, `normal_shift_norm_bound`
Theorem 8.1	`PoissonQuarticHolderEverywhere.no_local_holder_at_datum`
Corollary 8.2	`BanachFiniteDirectionStability.poisson_constant_finite_directions`, `no_uniform_one_direction_bound`
Proposition 9.1	`BanachQuarticDisjointObstruction.no_local_linear_bound_at_disjoint_image`

The complete-space namespace is essential in the existence row: an earlier namespace contains only a finite-dimensional existence theorem, although its energy derivative is dimension independent. The map called `normalFourEquiv` is a nonlinear bijection, not a linear equivalence or an identification supplied by a Hilbert inner product. The paper's H^* is the continuous real dual.

The baseline audit receipt records a successful complete root build, 54,943 audited declarations, and acceptance only of `propext`, `Classical.choice`, and `Quot.sound`. The editorial work changes no Lean source or dependency configuration. It reuses that existing verification rather than claiming a new full build. The source hashes and receipt provenance are included with the manuscript. Lean is pinned to version 4.34.1 and Mathlib to revision `d13f23b723b8a846827a245b89c10fc7d3f11612`.

11 What the estimates do and do not decide

The lower bound on δ gives existence on the whole continuous dual, uniqueness, and a global modulus. A weighted lower bound gives stronger control relative to one source. Nonvanishing almost everywhere implies injectivity of the weighted map; finite dimension converts that injectivity into a lower bound. Infinite Poisson space shows exactly where that last implication fails, and direct perturbations show that the failure corresponds to an actual inverse singularity.

No stochastic integral is constructed by these arguments. In particular, a Banach-space lower bound for a proposed stochastic divergence cannot be presumed from the existence theorem: it is an input that must be established for the actual operator. The results also do not settle general exponents, unbounded transpose domains, or closability characterizations. These are separate questions.

The solvability and uniform monotonicity belong to established nonlinear analysis. The bounded preliminary literature check places them in that context and does not certify priority for the particular combination of atomic optimality and finite-direction regularity. This manuscript is a reviewed exposition of verified mathematics, with mathematical novelty explicitly unresolved. An external submission would require a more targeted comparison with the literature on inverse duality mappings and their local moduli.

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